Cell lines immunopeptidome analysis - Fig2 - Fig S2

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Load packages

```r
library(ggplot2)
library(ggseqlogo)
library(reshape2)
library(UniProt.ws)
library(dplyr)
library(tidyr)
library(VennDiagram)
```

Load data

```r
# Cell lines immunopeptidomic
df_ch_mhc1 <- read.table("../data/immunopep_CH2879.tsv", sep = "\t",
    header = T)
df_jj_mhc1 <- read.table("../data/immunopep_JJ012.tsv", sep = "\t",
    header = T)

# Read IEDB file
df_iedb_mhc1 <- read.csv("../data/epitope_table_export_human_MHC_I.tsv",
    header = T, sep = "\t")

# CTA
l_CTA <- read.table("../data/CTA_list_clean.txt")$V1

# Read and prepare strong binders files
df_sb_jj <- read.table("../data/table_res_netmhc_all_sb_JJ012.tsv",
    sep = "\t", header = T)
df_sb_jj <- df_sb_jj[, c("Allele", "Peptide")]
df_sb_ch <- read.table("../data/table_res_netmhc_all_sb_CH2879.tsv",
    sep = "\t", header = T)
df_sb_ch <- df_sb_ch[, c("Allele", "Peptide")]

# Read RNAseq data
df_ch2879_tpm <- read.table("../results/CH2879_fpkm_tpm.tsv",
    sep = "\t", header = T)
df_jj012_tpm <- read.table("../results/JJ012_fpkm_tpm.tsv", sep = "\t",
    header = T)

# Microarray data
df_chs <- read.table("../results/whole_gene_int_CTA_sign_imm_clean.tsv",
    header = T, sep = "\t", row.names = 1)
df_chs <- df_chs[, -c(1, 2)]
df_metadata <- read.table("../results/metadata.tsv", header = T,
    sep = "\t")
```

I. Sequence length histogram

```
# Identify categories
seq_unique_ch <- setdiff(df_ch_mhc1$Seq, df_jj_mhc1$Seq)
seq_unique_jj <- setdiff(df_jj_mhc1$Seq, df_ch_mhc1$Seq)
seq_common <- intersect(df_ch_mhc1$Seq, df_jj_mhc1$Seq)

# Add category column
df_ch_mhc1$Category <- ifelse(df_ch_mhc1$Seq %in% seq_common,
    "Common peptides", ifelse(df_ch_mhc1$Seq %in% seq_unique_ch,
        "CH2879", NA))
df_jj_mhc1$Category <- ifelse(df_jj_mhc1$Seq %in% seq_common,
    "Common peptides", ifelse(df_jj_mhc1$Seq %in% seq_unique_jj,
        "JJ012", NA))

# Merge
pep_all <- bind_rows(df_ch_mhc1, df_jj_mhc1)

# Order colors
pep_all$Category <- factor(pep_all$Category, levels = c("CH2879",
    "JJ012", "Common peptides"))

# Histogram svg('../results/barplot_length_common.svg')
ggplot(pep_all, aes(x = seq_len, fill = Category)) + geom_histogram(binwidth = 1,
    position = "stack") + scale_fill_manual(values = c(CH2879 = "#000000ff",
    JJ012 = "#625f6aff", `Common peptides` = "#9e9e9eff")) +
    labs(x = "Peptide length (AA)", y = "Peptide number", fill = "Cell lines") +
    theme_minimal()
```

```
# dev.off()
```

I. Venn diagram MHCI

```
# IEDB pep
df_pep_mhc1 <- df_iedb_mhc1[, c("Name", "Source.Molecule.IRI",
    "Molecule.Parent.IRI")]
pep_iedb_mhc1 <- unique(df_pep_mhc1$Name)
pep_iedb_mhc1 <- gsub(" .*|\\+.*", "", pep_iedb_mhc1)
pep_iedb_mhc1 <- unique(pep_iedb_mhc1)

# Filter on length
df_jj_mhc1 <- df_jj_mhc1[df_jj_mhc1$seq_len >= 8 & df_jj_mhc1$seq_len <=
    11, ]
df_ch_mhc1 <- df_ch_mhc1[df_ch_mhc1$seq_len >= 8 & df_ch_mhc1$seq_len <=
    11, ]

# Delete known seq
new_pep_ch_mhc1 <- setdiff(df_ch_mhc1$Seq, pep_iedb_mhc1)
grep_ch <- sapply(new_pep_ch_mhc1, function(x) any(grepl(x, pep_iedb_mhc1)))
```

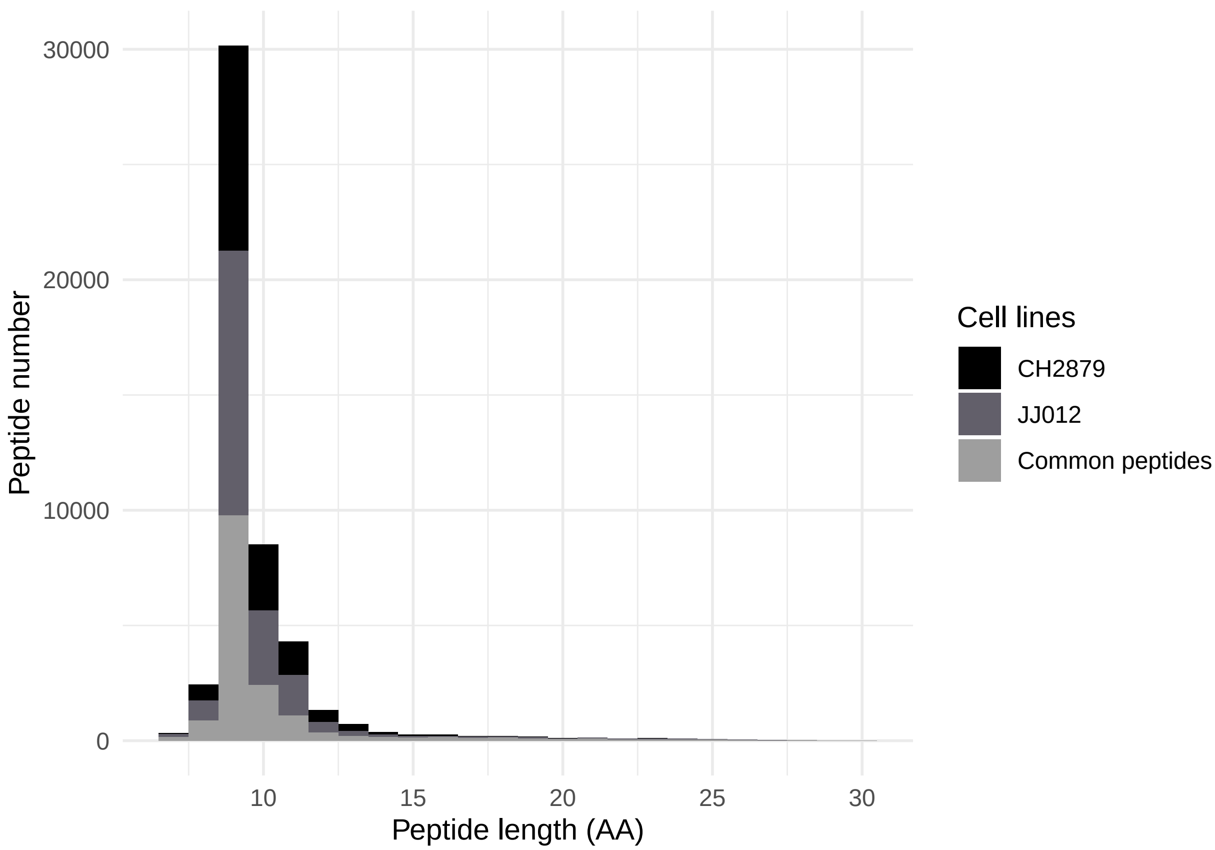

Figure 1: Peptides length histogram (MHC-I)

```r
new_pep_ch_mhc1 <- new_pep_ch_mhc1[!new_pep_ch_mhc1 %in% names(grep_ch)[grep_ch]]

# Select new pep JJ
new_pep_jj_mhc1 <- setdiff(df_jj_mhc1$Seq, pep_iedb_mhc1)
grep_jj <- sapply(new_pep_jj_mhc1, function(x) any(grepl(x, pep_iedb_mhc1)))
new_pep_jj_mhc1 <- new_pep_jj_mhc1[!new_pep_jj_mhc1 %in% names(grep_jj)[grep_jj]]

# Clean IEDB pep
pep_iedb_extended_ch <- unique(c(pep_iedb_mhc1, names(grep_ch)[grep_ch]))
pep_iedb_extended_jj <- unique(c(pep_iedb_mhc1, names(grep_jj)[grep_jj]))
```

JJ012

```r
# Venn diagram
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_jj),
    area2 = length(df_jj_mhc1$Seq), cross.area = length(df_jj_mhc1$Seq) -
        length(new_pep_jj_mhc1), category = c("IEDB", "JJ012"),
    fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

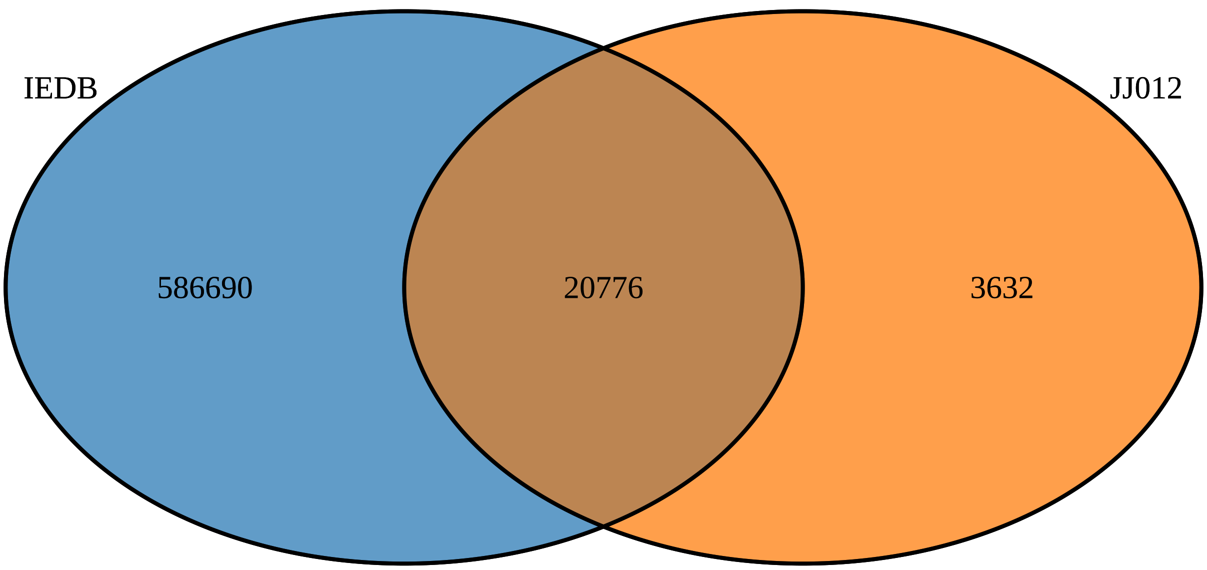

Figure 2: Venn plot JJ012 (MHCI)

CH2879

```r
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_ch),
    area2 = length(df_ch_mhc1$Seq), cross.area = length(df_ch_mhc1$Seq) -
        length(new_pep_ch_mhc1), category = c("IEDB", "CH2879"),
    fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

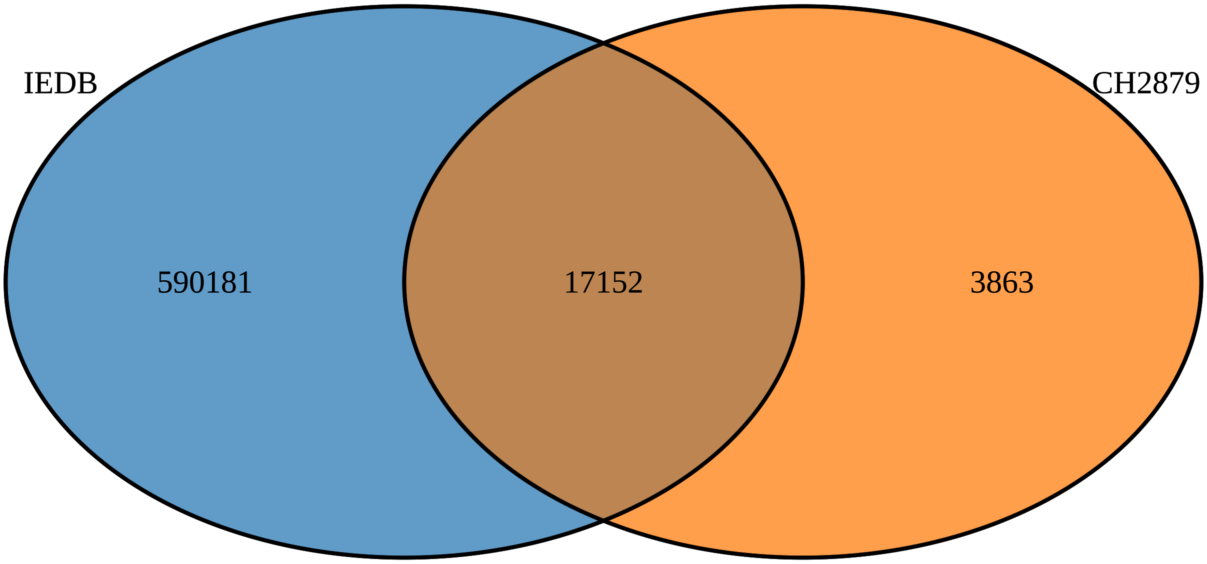

Figure 3: Venn plot CH2879 (MHCI)

II. Proportion of strong binders peptides per HLA-type

JJ012

```
# Prepare
df_sb_jj <- df_sb_jj[df_sb_jj$Peptide %in% df_jj_mhc1$Seq, ]
df_sb_jj$Category <- ifelse(df_sb_jj$Peptide %in% new_pep_jj_mhc1,
    "Novel peptide", "Known peptide")
df_sb_jj <- df_sb_jj %>%
    distinct(Peptide, .keep_all = TRUE)

# Compute percentages
df_percent <- df_sb_jj %>%
    group_by(Allele, Category) %>%
    summarise(n = n(), .groups = "drop") %>%
    mutate(percent = 100 * n/sum(n))

# Filter on HLA types
hla_jj <- c("HLA-A0201", "HLA-B4402", "HLA-C0501")
df_hla <- df_percent %>%
    filter(Allele %in% hla_jj)

# Plot svg('../results/barplot_category_pep_jj.svg')
ggplot(df_hla, aes(y = Allele, x = percent, fill = Category)) +
    geom_bar(stat = "identity", position = "stack", width = 0.7) +
    scale_fill_manual(values = c(`Known peptide` = "#2C7BB6",
        `Novel peptide` = "#ff7f0e")) + labs(x = "% of JJ012 peptides (strong binders) per HLA type",
    y = NULL, fill = NULL) + theme_classic()
```

```
# dev.off()
```

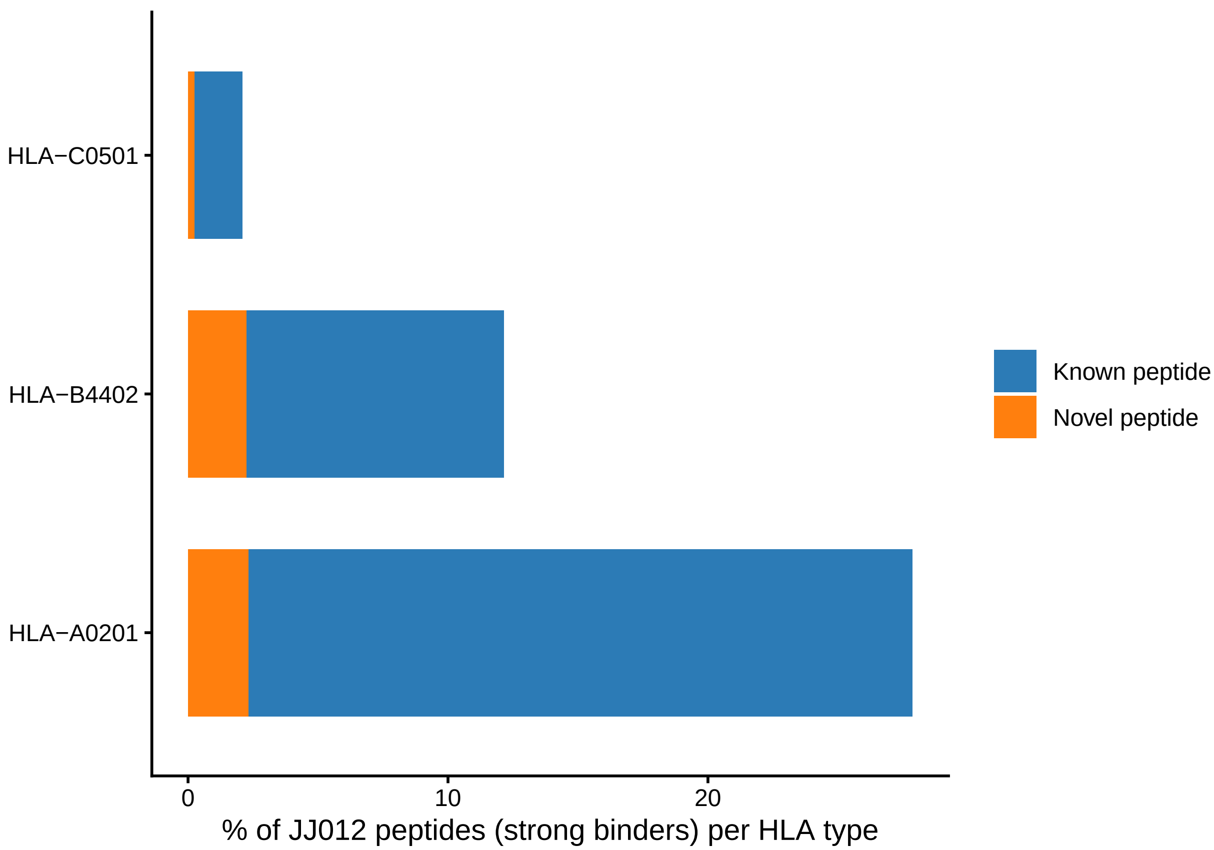

Figure 4: Histogram of strong binders peptides per HLA-type (JJ012)

CH2879

```r
# Prepare
df_sb_ch <- df_sb_ch[df_sb_ch$Peptide %in% df_ch_mhc1$Seq, ]
df_sb_ch$Category <- ifelse(df_sb_ch$Peptide %in% new_pep_ch_mhc1,
    "Novel peptide", "Known peptide")
df_sb_ch <- df_sb_ch %>%
    distinct(Peptide, .keep_all = TRUE)

# Compute percentages
df_percent <- df_sb_ch %>%
    group_by(Allele, Category) %>%
    summarise(n = n(), .groups = "drop") %>%
    mutate(percent = 100 * n/sum(n))

# Filter on HLA types
hla_ch <- c("HLA-A0101", "HLA-B0702", "HLA-C0702")
df_hla <- df_percent %>%
    filter(Allele %in% hla_ch)

# Plot svg('../results/barplot_category_pep_ch.svg')
ggplot(df_hla, aes(y = Allele, x = percent, fill = Category)) +
    geom_bar(stat = "identity", position = "stack", width = 0.7) +
    scale_fill_manual(values = c(`Known peptide` = "#2C7BB6",
        `Novel peptide` = "#ff7f0e")) + labs(x = "% of CH2879 peptides (strong binders) per HLA type",
    y = NULL, fill = "Category") + theme_classic()
```

```r
# dev.off()
```

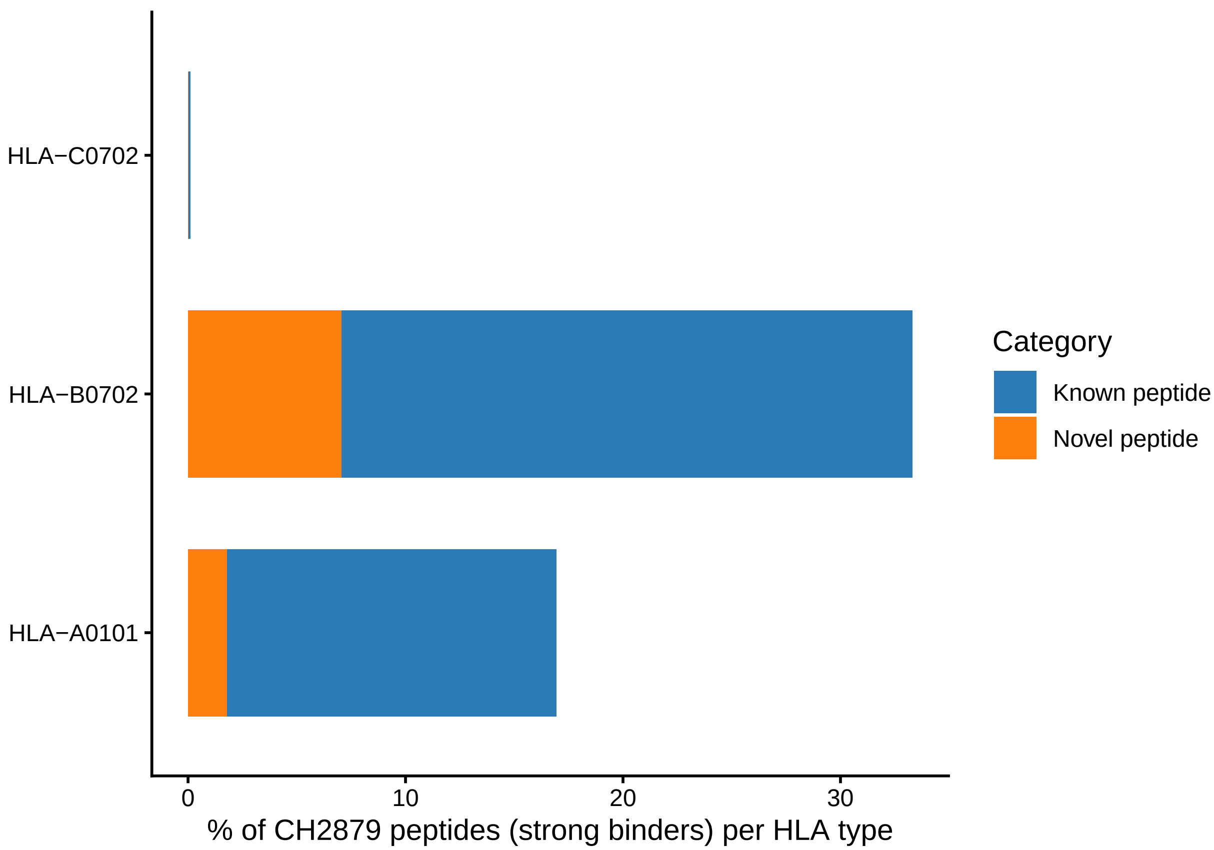

Figure 5: Histogram of strong binders peptides per HLA-type (CH2879)

III. Sequence logo for 9-mer peptides for SB per HLA type

JJ012

```r
# Take peptide length = 9 AA
df <- df_sb_jj %>%
    filter(Allele %in% hla_jj) %>%
    filter(nchar(Peptide) == 9)
seqs_by_hla <- split(df$Peptide, df$Allele)

# Count peptides per group
counts <- lengths(seqs_by_hla)

# Rename to have the number per HLA
names(seqs_by_hla) <- paste0(names(seqs_by_hla), " (n=", counts,
    ")")

# svg('../results/seq_logos_jj.svg')
ggseqlogo(seqs_by_hla, ncol = 3, seq_type = "aa") + theme_bw() +
    theme(plot.title = element_text(hjust = 0.5), panel.grid.major = element_line(color = "grey80"),
        panel.grid.minor = element_blank()) + xlab("Position") +
    ylab("Information (bits)")
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the ggseqlogo package.
##   Please report the issue at <https://github.com/omarwagih/ggseqlogo/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
# dev.off()
```

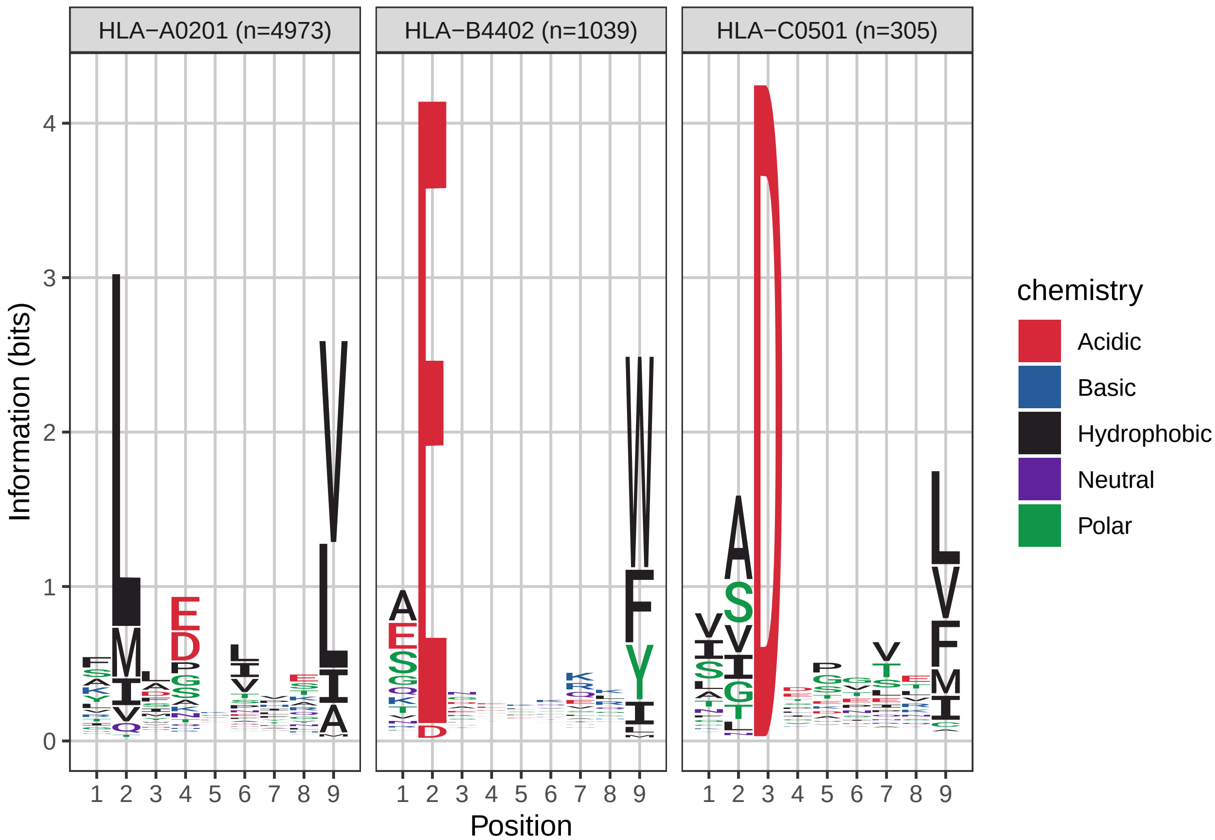

Figure 6: Sequence logo for SB per HLA type (JJ012)

CH2879

```
# Take peptide length = 9 AA
df <- df_sb_ch %>%
    filter(Allele %in% hla_ch) %>%
    filter(nchar(Peptide) == 9)
seqs_by_hla <- split(df$Peptide, df$Allele)

# Count peptides per group
counts <- lengths(seqs_by_hla)

# Rename to have the number per HLA
names(seqs_by_hla) <- paste0(names(seqs_by_hla), " (n=", counts,
    ")")

# svg('../results/seq_logos_ch.svg')
ggseqlogo(seqs_by_hla, ncol = 3, seq_type = "aa") + theme_bw() +
    theme(plot.title = element_text(hjust = 0.5), panel.grid.major = element_line(color = "grey80"),
        panel.grid.minor = element_blank()) + xlab("Position") +
    ylab("Information (bits)")
```

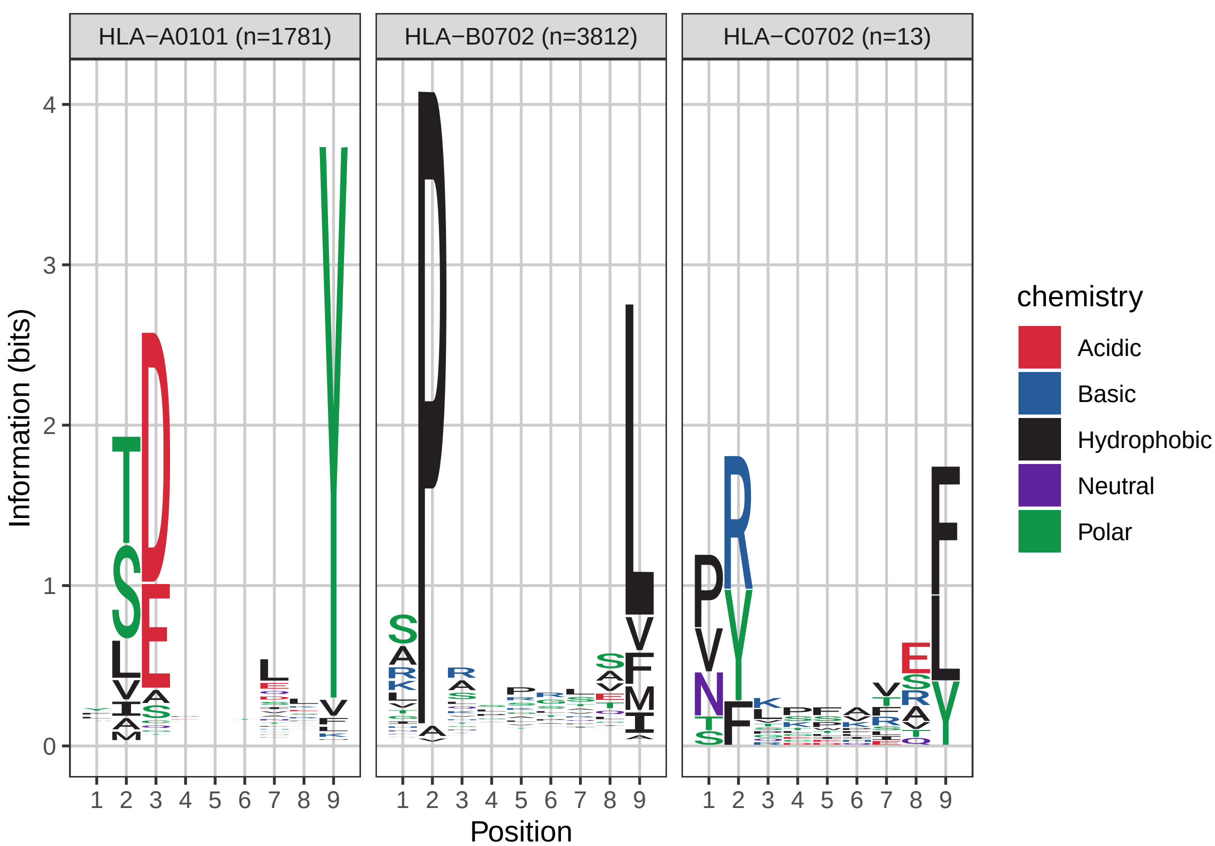

Figure 7: Sequence logo for SB per HLA type (CH2879)

```r
# dev.off()
```

IV. FigS2, correlation between RNA from cell lines and patients (microarray)

```
# Clean data
df_ch2879_tpm <- df_ch2879_tpm[, c("Gene_ID", "Gene_Name", "TPM")]
df_jj012_tpm <- df_jj012_tpm[, c("Gene_ID", "Gene_Name", "TPM")]

# Rename
colnames(df_ch2879_tpm)[3] <- "TPM_CH2879"
colnames(df_jj012_tpm)[3] <- "TPM_JJ012"

# Merge
df_merge <- merge(df_ch2879_tpm, df_jj012_tpm, by = c("Gene_ID",
    "Gene_Name"), all = TRUE)

df_all <- df_merge[!df_merge$Gene_ID %in% c("ENSG00000233098",
    "ENSG00000291074", "ENSG00000293247", "ENSG00000293226"),
    ]

# Take CTAs
df_cta <- df_all[df_all$Gene_Name %in% l_CTA, ]
rownames(df_cta) <- df_cta$Gene_Name
df_cta <- df_cta[, c(3, 4)]
```

```
# Select common genes
common_CTA <- intersect(rownames(df_cta), rownames(df_chs))
common_CTA <- common_CTA[common_CTA %in% l_CTA]
df_cells_CTA <- df_cta[common_CTA, c("TPM_CH2879", "TPM_JJ012")]
df_chs_CTA <- df_chs[common_CTA, ]

# Take corresponding grades
df_chs_CTA_g2 <- df_chs[common_CTA, colnames(df_chs) %in% df_metadata$Patient[df_metadata$Histology ==
    "G2"]]
df_chs_CTA_g3 <- df_chs[common_CTA, colnames(df_chs) %in% df_metadata$Patient[df_metadata$Histology ==
    "G3"]]

# Compute means
chs_mean_CTA_g2 <- rowMeans(df_chs_CTA_g2)
chs_mean_CTA_g3 <- rowMeans(df_chs_CTA_g3)

# Log2 transformation
df_cells_log2_CTA <- log2(df_cells_CTA + 1)
```

```
# Correlation CH2879 vs CHS (CTA)
cor_CH2879_CTA <- cor.test(df_cells_log2_CTA$TPM_CH2879, chs_mean_CTA_g3,
    method = "spearman")
```

```
## Warning in cor.test.default(df_cells_log2_CTA$TPM_CH2879, chs_mean_CTA_g3, :
## Impossible de calculer la p-value exacte avec des ex-aequos
```

```r
# Correlation JJ012 vs CHS (CTA)
cor_JJ012_CTA <- cor.test(df_cells_log2_CTA$TPM_JJ012, chs_mean_CTA_g2,
    method = "spearman")
```

```
## Warning in cor.test.default(df_cells_log2_CTA$TPM_JJ012, chs_mean_CTA_g2, :
## Impossible de calculer la p-value exacte avec des ex-aequos
```

```r
# Results
cor_CH2879_CTA$estimate
```

```
##       rho
## 0.4135692
```

```r
cor_CH2879_CTA$p.value
```

```
## [1] 3.663447e-36
```

```r
cor_JJ012_CTA$estimate
```

```
##       rho
## 0.3266071
```

```r
cor_JJ012_CTA$p.value
```

```
## [1] 2.095259e-22
```

```r
df_plot_CTA_CH2879 <- data.frame(CHS = chs_mean_CTA_g3, CH2879 = df_cells_log2_CTA$TPM_CH2879)

# Plot
p <- ggplot(df_plot_CTA_CH2879, aes(x = CHS, y = CH2879)) + geom_point(size = 2) +
    geom_smooth(method = "lm", se = FALSE, color = "blue") +
    annotate("text", x = max(df_plot_CTA_CH2879$CHS), y = max(df_plot_CTA_CH2879$CH2879),
        label = paste0("rho = ", round(cor_CH2879_CTA$estimate,
            2), "\np = ", signif(cor_CH2879_CTA$p.value, 3)),
        hjust = 1, vjust = 1, size = 4) + ggtitle("Correlation CHS G3 vs CH2879 (CTA genes)") +
    xlab("CHS grade 3 patients") + ylab("CH2879 TPM (log2 + 1)") +
    theme_classic()
p
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```r
df_plot_CTA_JJ012 <- data.frame(CHS = chs_mean_CTA_g2, JJ012 = df_cells_log2_CTA$TPM_JJ012)

# Plot
p <- ggplot(df_plot_CTA_JJ012, aes(x = CHS, y = JJ012)) + geom_point(size = 2) +
    geom_smooth(method = "lm", se = FALSE, color = "blue") +
    annotate("text", x = max(df_plot_CTA_JJ012$CHS), y = max(df_plot_CTA_JJ012$JJ012),
        label = paste0("rho = ", round(cor_JJ012_CTA$estimate,
```

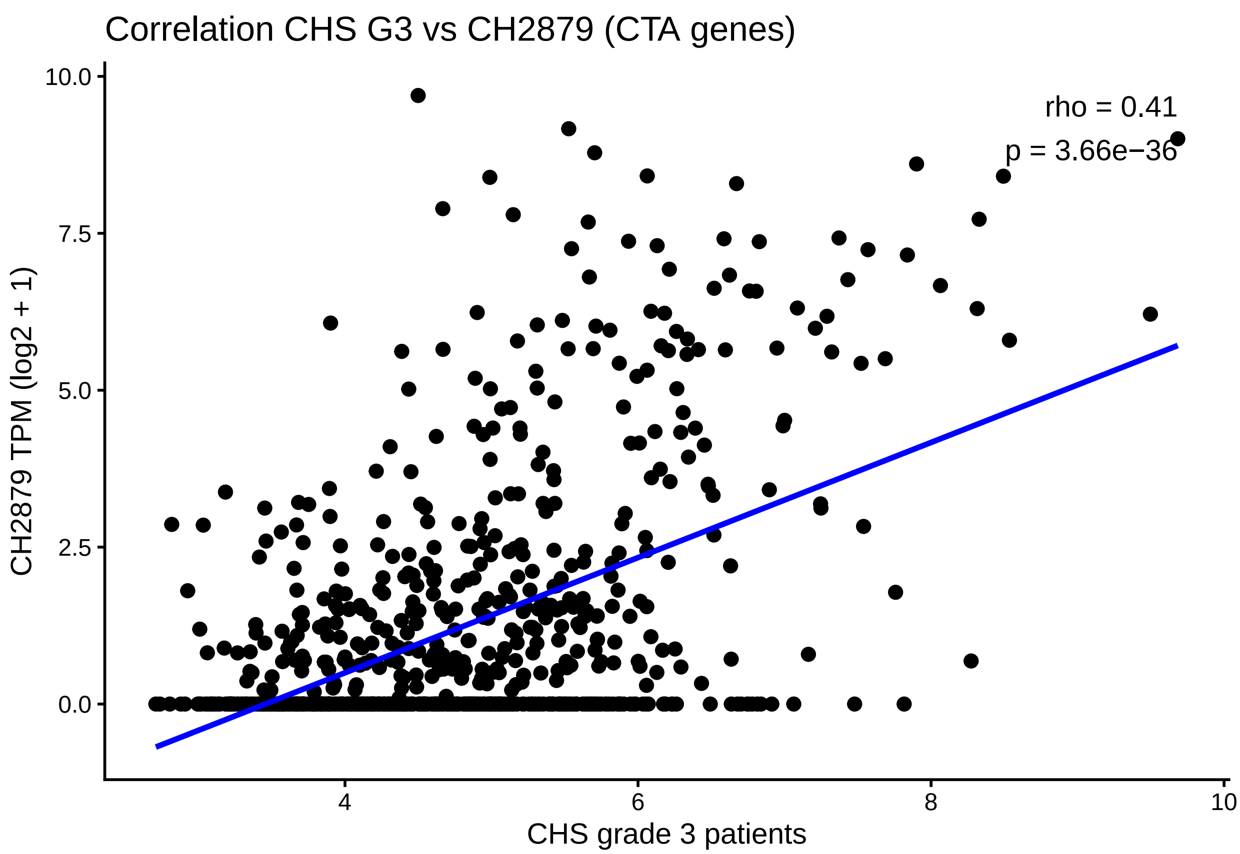

Figure 8: Correlation plot between g3 and CH

```r
            2), "\np = ", signif(cor_JJ012_CTA$p.value, 3)),
        hjust = 1, vjust = 1, size = 4) + ggtitle("Correlation CHS G2 vs JJ012 (CTA genes)") +
    xlab("CHS grade 2 patients") + ylab("JJ012 TPM (log2 + 1)") +
    theme_classic()
p
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

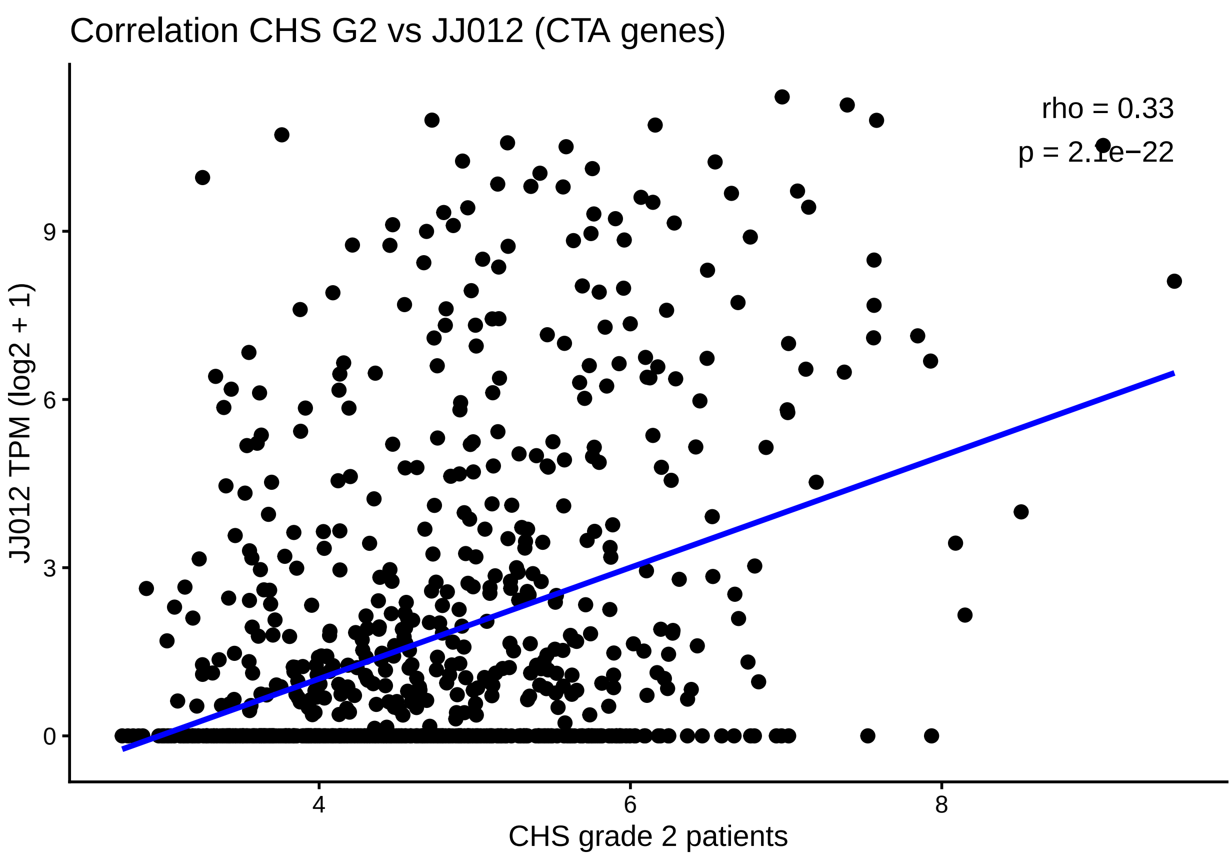

Figure 9: Correlation plot between g2 and JJ